un controlled

single - phase

Half wave

→ $V_{dc} = \frac{V_m}{\pi}$

→ $I_{dc} = \frac{V_m}{\pi R}$

→ $V_{rms} = \frac{V_m}{2}$

→ $I_{rms} = \frac{V_m}{2R}$

→ $P_{dc} = \frac{V_m^2}{\pi^2 R}$

→ $P_{ac} = \frac{V_m^2}{4R}$

→ $\eta = 40.5\ \%$

→ $R.F = \frac{V_{RI}(V_{ac})}{V_{dc}} = 1.2114$

→ $F.F = \frac{V_{rms}}{V_{dc}} = 1.5707$

→ $V_{RI} = V_{ac} = \sqrt{V_{rms}^2 - V_{dc}^2}$

full wave

→ $V_{dc} = \frac{2V_m}{\pi}$

→ $I_{dc} = \frac{2V_m}{\pi R}$

→ $V_{rms} = \frac{V_m}{\sqrt{2}}$

→ $I_{rms} = \frac{V_m}{\sqrt{2}\ R}$

→ $P_{dc} = \frac{4\ V_m^2}{\pi^2 R}$

→ $P_{ac} = \frac{V_m^2}{2\ R}$

→ $\eta = 81.05\ \%$

→ $R.F = 0.48343$

→ $F.F = 1.1107$

controlled

with α

Single - phase

Half wave

→ $V_{dc} = \frac{V_m}{2\pi}(1+\cos\alpha)$

→ $I_{dc} = \frac{V_m}{2\pi R}(1+\cos\alpha)$

→ $V_{rms} = \frac{V_m}{2}\sqrt{\frac{1}{\pi}\left(\pi-\alpha+\frac{\sin 2\alpha}{2}\right)}$

→ $I_{rms} = \frac{V_m}{2R}\sqrt{\frac{1}{\pi}\left(\pi-\alpha+\frac{\sin 2\alpha}{2}\right)}$

→ $P_{ac} = \frac{V_{rms}^2}{R}$

→ $P_{dc} = \frac{V_{dc}^2}{R}$

Full wave

→ $V_{dc} = \frac{V_m}{\pi}(1+\cos\alpha)$

→ $I_{dc} = \frac{V_m}{\pi R}(1+\cos\alpha)$

→ $V_{rms} = V_m\sqrt{\frac{1}{2}-\frac{\alpha}{2\pi}+\frac{\sin 2\alpha}{4\pi}}$

→ $I_{rms} = \frac{V_m}{R}\sqrt{\frac{1}{2}-\frac{\alpha}{2\pi}+\frac{\sin 2\alpha}{4\pi}}$

→ $P_{ac} = \frac{V_{rms}^2}{R}$

→ $P_{dc} = \frac{V_{dc}^2}{R}$

* Single - phase half wave : (1 pulse)
* single - phase full wave : (2 pulse)

uncontrolled

Three-phase

Half wave

$\rightarrow V_{dc} = \dfrac{3\sqrt{3}\ V_m}{2\pi}$

$\rightarrow I_{dc} = \dfrac{3\sqrt{3}\ V_m}{2\pi R}$

$\rightarrow V_{rms} = 0.84 \cdot V_m$

$\rightarrow I_{rms} = \dfrac{0.84\ V_m}{R}$

$\rightarrow P_{dc} = \dfrac{27\ V_m^2}{4\pi^2 R}$

$\rightarrow P_{ac} = \dfrac{0.7056\ V_m^2}{R}$

Full wave

$\rightarrow V_{dc} = \dfrac{3\sqrt{3}\ V_m}{\pi}$

$\rightarrow I_{dc} = \dfrac{3\sqrt{3}\ V_m}{\pi R}$

$\rightarrow V_{rms} = 1.655\ V_m$

$\rightarrow I_{rms} = 0.78\ I_m$ (phase)

$\rightarrow I_{rms} = 0.552\ I_m$ (Diode)

$\hookrightarrow I_m = \dfrac{1.73\ V_m}{R}$

$\rightarrow P_{dc} = \dfrac{27\ V_m^2}{\pi^2 R}$

$\rightarrow P_{ac} = \dfrac{1.29\ V_m^2}{R}$

Controlled with α

Three-phase

Half wave

when $\alpha \leq 30°$

$$V_{dc} = \frac{3\sqrt{3}\,V_m}{2\pi}\cos\alpha$$

$$I_{dc} = \frac{3\sqrt{3}\,V_m}{2\pi R}\cos\alpha$$

$$V_{rms} = \sqrt{3}\,V_m\sqrt{\frac{1}{6}+\frac{\sqrt{3}}{8\pi}\cos 2\alpha}$$

$$I_{rms} = \frac{\sqrt{3}\,V_m}{R}\sqrt{\frac{1}{6}+\frac{\sqrt{3}}{8\pi}\cos 2\alpha}$$

when $\alpha \geq 30$

$$V_{dc} = \frac{3V_m}{2\pi}\left(1+\cos\left(\frac{\pi}{6}+\alpha\right)\right)$$

$$I_{dc} = \frac{3V_m}{2\pi R}\left(1+\cos\left(\frac{\pi}{6}+\alpha\right)\right)$$

$$V_{rms} = V_m\sqrt{\frac{3}{4\pi}\left[\frac{5\pi}{6}-\alpha+\frac{1}{2}\sin\left(\frac{\pi}{3}+2\alpha\right)\right]}$$

$$I_{rms} = \frac{V_m}{R}\sqrt{\frac{3}{4\pi}\left[\frac{5\pi}{6}-\alpha+\frac{1}{2}\sin\left(\frac{\pi}{3}+2\alpha\right)\right]}$$

Full wave

when $\alpha \leq 30$

$$V_{dc} = \frac{3\sqrt{3}\,V_m}{\pi}\cos\alpha$$

$$I_{dc} = \frac{3\sqrt{3}\,V_m}{\pi R}\cos\alpha$$

$$V_{rms} = \sqrt{3}\,V_m\sqrt{\frac{1}{2}+\frac{3\sqrt{3}}{4\pi}\cos 2\alpha}$$

$$I_{rms} = \frac{\sqrt{3}\,V_m}{R}\sqrt{\frac{1}{2}+\frac{3\sqrt{3}}{4\pi}\cos 2\alpha}$$

when $\alpha > 60$

$$V_{dc} = \frac{3\sqrt{3}\,V_m}{\pi}\cos\left(\frac{\pi}{3}+\alpha\right)$$

$$I_{dc} = \frac{3\sqrt{3}\,V_m}{\pi R}\cos\left(\frac{\pi}{3}+\alpha\right)$$

* Three-phase Half wave : (3 pulse)
* Three-phase full wave : (6 pulse)